Date :

Assignment 1

The difference of two convex functions is not necessarily convex

$f : \mathbb{R}^n \longrightarrow \mathbb{R}$ is said to be convex if $\forall x, y \in \mathbb{R}^n$ & $\forall \lambda \in [0,1]$

$$f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y)$$

Let f, g be convex functions

$\therefore \forall x, y \in \mathbb{R}^n$ & $\lambda, \mu \in [0,1]$

$$f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y)$$

&

$$g(\mu x + (1-\mu) y) \leq \mu g(x) + (1-\mu) g(y)$$

Suppose $f-g$ were convex

$$(f-g)(\lambda x + (1-\lambda) y) \leq \lambda (f-g)(x) + (1-\lambda)(f-g)(y)$$

$$f(\lambda x + (1-\lambda) y) - g(\lambda x + (1-\lambda) y) \leq \lambda f(x) - \lambda g(x) + (1-\lambda) f(y) - (1-\lambda) g(y)$$

Let $f(\lambda x + (1-\lambda) y) = \lambda f(x) + (1-\lambda) f(y) + c$

for some $c \geq 0$

for some fixed x, y

$$\Rightarrow c \leq g(\lambda x + (1-\lambda) y) - \lambda g(x) - (1-\lambda) g(y)$$

$$\Rightarrow g(\lambda x + (1-\lambda) y) \geq \lambda g(x) + (1-\lambda) g(y)$$

This gives a contradiction. $\therefore f-g$ is not necessarily convex